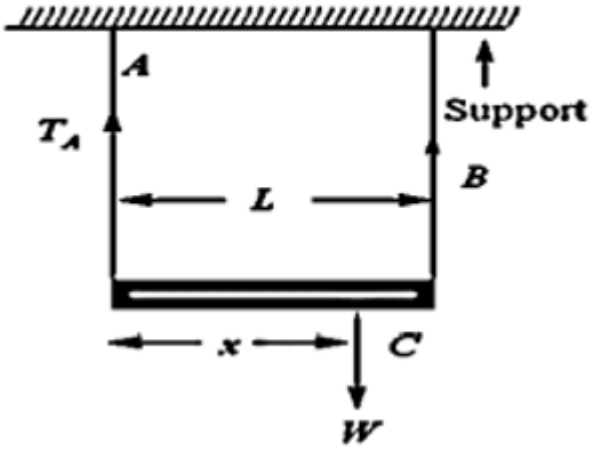
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A light rod of length L is suspended from a support horizontally by means of two vertical wires A and B of equal length as shown in figure. The cross sectional area of A is half that of B and the Young's modulus of A is twice that of B. A weight W is hung as shown. The value of x so that W produces equal stress in wires A and B is



(A) $\frac{L}{3}$

(B) $\frac{L}{2}$

(C) $\frac{2L}{3}$

(D) $\frac{3L}{4}$

Q.2

A steel wire of uniform cross-section of $2\ \mathrm{mm}^2$ is heated upto $50^{\circ}\mathrm{C}$ and clamped rigidly at two ends. If the temperature of wire falls to 30° then change in tension in the wire will be, if coefficient of linear expansion of steel is $1.1 \times 10^{-5}/^{\circ}\mathrm{C}$ and young's modulus of elasticity of steel is $2 \times 10^{11}\ \mathrm{N/m}^2$.

(A) $44\ \mathrm{N}$

(B) $88\ \mathrm{N}$

(C) $132\ \mathrm{N}$

(D) $22\ \mathrm{N}$

Q.3

A uniform copper rod of length ' Γ density ' ρ ' with ultimate tensile strength ' σ ' is rotatedin a horizontal plane about an axis perpendicular to length and passing through itscentre. The frequency of rotation (in revolutions per sec) at which the rod ruptures is

(A) $\sqrt{\frac{\sigma}{\rho}\frac{1}{\pi l}}$

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(B) $\sqrt{\frac{\sigma}{2\rho}}\frac{1}{\pi l}$

(C) $\sqrt{\frac{2\sigma}{\rho}}\frac{1}{\pi l}$

(D) $\sqrt{\frac{\sigma}{2\rho}}\frac{1}{2\pi l}$

Q.4

A spring is stretched by $3\ \text{cm}$ when a load of 5.4×10^6 dyne is suspended from it. Work done will be-

(A) $8.1 \times 10^6\ \text{J}$

(B) $8 \times 10^6\ \text{J}$

(C) 8.0×10^6 ergs

(D) 8.1×10^6 ergs

Q.5

When the tension in a metal wire is T_1, its length is I_1. When the tension is T_2, its length I_2. The natural length of wire is

(A) $\frac{T_2}{T_1}(I_1 + I_2)$

(B) $T_{I_1} + I_2 l_2$

(C) $\frac{I_1 T_2 - I_2 T_1}{T_2 - T_1}$

(D) $\frac{I_1 T_2 + I_2 T_1}{T_2 + T_1}$

Q.6

The volume of a solid rubber ball (C) when it is carried from the surface to the bottom of a $200\ \text{m}$ deep lake decrease by 0.1%. The value for bulk modulus elasticity for rubber will be

(A) $2 \times 10^9\ \text{Pa}$

(B) $2 \times 10^6\ \text{Pa}$

(C) $2 \times 10^4\ \text{Pa}$

(D) $2 \times 10^{-4}\ \text{Pa}$

Q.7

A thin ring has a radius R, density ρ and Young's modulus Y. The ring is rotated in its own plane about an axis passing through its centre with angular velocity ω. Then the small increase in its radius is :

(A) $dR = \frac{\rho\omega^2 R^3}{Y}$

(B) $dR = \frac{3\rho\omega^2 R^3}{Y}$

(C) $\mathrm{dR} = \frac{6\rho\omega^2 \mathrm{R}^3}{\mathrm{Y}}$

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(D) $dR = \frac{\rho\omega^2 R^3}{2Y}$

Q.8

A brass rod is to support a load of $400\ \text{N}$. If its elastic limit is $4.0 \times 10^8\ \text{N/m}^2$ its minimum diameter must be -

(A) $1.13\ \text{mm}$

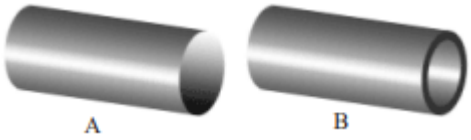
(B) $2.26\ \text{mm}$

(C) $3.71\ \text{mm}$

(D) $4.52\ \text{mm}$

Q.9

Figures A and B show two cylinders that are identical in all respects, except one is hollow. Identical forces are applied to each cylinder in order to stretch them. Choose the correct options -



(A) A and B both stretch by the same amount.

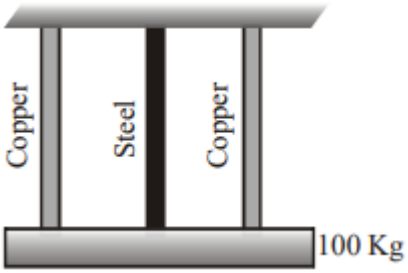
(B) A stretches more than B.

(C) B stretches more than A.

(D) None of these

Q.10

A heavy plank of mass $100\ \text{kg}$ hangs on three vertical wires of equal length arranged symmetrically (see figure). All the wires have the same cross section. The middle wire is of steel and the other two are of copper. The modulus of elasticity of steel is assumed to be double that of copper. Choose the correct options for the tensions in the wire.



(A) $\text{T}_{\text{cu}} = 245\ \text{N}$

(B) $\text{T}_{\text{steel}} = 490\ \text{N}$

(C) $\text{T}_{\text{cu}} = 215\ \text{N}$

(D) $\text{T}_{\text{steel}} = 470\ \text{N}$

Q.11

Which among the following statement is/are correct -

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(A) If a heavy uniform rod is suspended vertically, its diameter is largest at the bottom and smallest at the top.

(B) If a uniform rod is fixed at its ends and heated, the force developed in the rod is not dependent on its length.

(C) The shear modulus of a liquid is infinite.

(D) Bulk modulus of a perfectly rigid body is zero.

Q.12

Passage:
A body of mass m is attached to the lower end of a vertically suspended wire. The elongation in the wire is ℓ.
Question: Loss in gravitational potential energy is -

(A) $\mathrm{mg}\ell$

(B) $mg\ell/2$

(C) $2\mathrm{mg}\ell$

(D) $\mathrm{mg}\ell/3$

Q.13

Passage:
A body of mass m is attached to the lower end of a vertically suspended wire. The elongation in the wire is ℓ.
Question: Gain in elastic P.E. is -

(A) $\mathrm{mg}\ell$

(B) $\mathrm{mg}\ell/2$

(C) $2\mathrm{mg}\ell$

(D) $\mathrm{mg}\ell/3$

Q.14

Passage:
A body of mass m is attached to the lower end of a vertically suspended wire. The elongation in the wire is ℓ.
Question: Heat produced equal to -

(A) $\mathrm{mg}\ell$

(B) $\mathrm{mg}\ell/2$

(C) $2\mathrm{mg}\ell$

(D) $mg\ell/3$

Q.15

Passage:
Young's Modulus of elasticity (Y) is defined as the ratio of normal stress to the longitudinal strain with in the elastic limit.
Thermal stress depends on Young's Modulus of the material to understand it consider a bar which is heated & then

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prevented from contracting as it cools. A considerable force will be exerted at the ends of the bar, if the bar were free to contract, then decrease in length,

$\Delta\ell = \ell\alpha\theta \quad \ldots$ (i)

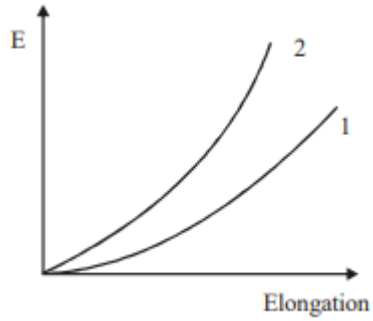
Where ℓ is the original length, α is the coefficient of linear expansion \& θ is the fall of temperature.

Again, $\mathrm{Y} = \frac{\mathrm{F/A}}{\Delta\ell/\ell} = \frac{\mathrm{F}\ell}{\mathrm{A}\Delta\ell}$ or $\Delta\ell = \frac{\mathrm{F}\ell}{\mathrm{AY}} \ldots (ii)$

Where, F is the force developed in the bar & A is the cross-sectional area of the bar.

From (i) & (ii), $\frac{\mathrm{F}\ell}{\mathrm{AY}} = \ell\alpha\theta$ or $\mathrm{F} = \mathrm{YA}\alpha\theta$

Again, thermal stress $= \frac{\mathrm{F}}{\mathrm{A}}$ or Thermal stress $= \mathrm{Y}\alpha\theta.$



Question: The plot of elastic strain energy of per unit volume vs elongation for two wires are as shown. If Y is Young's modulus, then -

(A) $\mathrm{Y}_1 > \mathrm{Y}_2$

(B) $Y_1 = Y_2$

(C) $\mathrm{Y}_1 < \mathrm{Y}_2$

(D) cannot be concluded

Q.16

Passage:

Young's Modulus of elasticity (Y) is defined as the ratio of normal stress to the longitudinal strain with in the elastic limit.

Thermal stress depends on Young's Modulus of the material to understand it consider a bar which is heated & then prevented from contracting as it cools. A considerable force will be exerted at the ends of the bar, if the bar were free to contract, then decrease in length,

$\Delta\ell = \ell\alpha\theta \quad \ldots$ (i)

Where ℓ is the original length, α is the coefficient of linear expansion & θ is the fall of temperature.

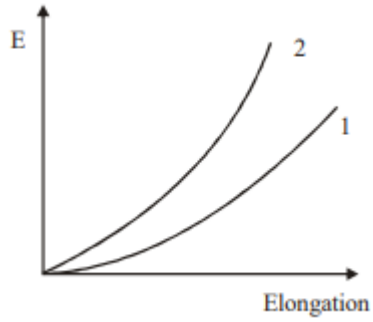
Again, $\mathrm{Y} = \frac{\mathrm{F/A}}{\Delta\ell/\ell} = \frac{\mathrm{F}\ell}{\mathrm{A}\Delta\ell}$ or $\Delta\ell = \frac{\mathrm{F}\ell}{\mathrm{AY}} \ldots (ii)$

Where, F is the force developed in the bar & A is the cross-sectional area of the bar.

From (i) & (ii), $\frac{\mathrm{F}\ell}{\mathrm{AY}} = \ell\alpha\theta$ or $\mathrm{F} = \mathrm{YA}\alpha\theta$

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Again, thermal stress $= \frac{F}{A}$ or Thermal stress $= Y\alpha\theta$.



Question: Two entirely different rods are joined end to end, to form a composite rod. When it was heated through $\Delta\theta$, its length increased by x. When it was subjected to a tensile stress σ, its length increased by y. When it was fixed between fixed supports and heated through $\Delta\theta$, the stress in the rod was less than σ. Then -

(A) $x > y$

(B) $x = y$

(C) $x < y$

(D) cannot be concluded with given data.

Q.17

Passage:

Young's Modulus of elasticity (Y) is defined as the ratio of normal stress to the longitudinal strain with in the elastic limit.

Thermal stress depends on Young's Modulus of the material to understand it consider a bar which is heated & then prevented from contracting as it cools. A considerable force will be exerted at the ends of the bar, if the bar were free to contract, then decrease in length,

$\Delta\ell = \ell\alpha\theta \quad \ldots$ (i)

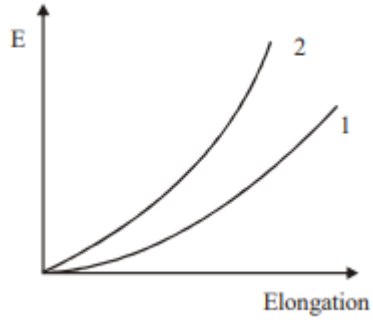
Where ℓ is the original length, α is the coefficient of linear expansion & θ is the fall of temperature.

Again, $Y = \frac{F/A}{\Delta\ell/\ell} = \frac{F\ell}{A\Delta\ell}$ or $\Delta\ell = \frac{F\ell}{AY} \ldots (ii)$

Where, F is the force developed in the bar & A is the cross-sectional area of the bar.

From (i) & (ii), $\frac{F\ell}{AY} = \ell\alpha\theta$ or $F = YA\alpha\theta$

Again, thermal stress $= \frac{F}{A}$ or Thermal stress $= Y\alpha\theta$.



Question: A horizontal beam of mass $1234\ \text{kg}$ is suspended by three wires of identical cross-sections, and each $1\ \text{m}$ long. One is steel (Young's modulus $20 \times 10^{10}\text{Nm}^{-2}$), another brass $\left(Y = 8 \times 10^{10}\text{Nm}^{-2}\right)$ the third

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made by joining steel and brass wires end to end. If the beam remains horizontal and if the maximum tension among the three is $6170\ \mathrm{N}$, determine the length of steel wire used in the composite wire. $\mathrm{g} = 10\ \mathrm{ms}^{-2}$.

(A) 0.56

(B) 0.12

(C) 0.78

(D) 0.92

Q.18

Match the column -

Column I	**Column II**
(A) The bulk modulus of water from the following data : Initial volume = 100.0 litre, Pressure increase = 100.0 atm. Final volume = 100.5 litre.	(p) $2.03 \times 10^9\ \mathrm{Nm}^{-2}$.
(B) A steel wire of length 4.7 m and cross section $3.0 \times 10^{-5}\ \mathrm{m}^2$ stretches by the same amount as a copper wire of length 3.5 m and cross section $4.0 \times 10^{-5}\ \mathrm{m}^2$ under a given load. The ratio of Young's modulus of steel to that of copper	(q) 1.79
(C) A material has Poisson's ratio 0.50. If a uniform rod of it suffers a longitudinal strain of 2×10^{-3}, then the percentage change in volume is	(r) 0.4
(D) If a rubber ball is taken at the depth of 200 m in a pool, its volume decreases by 0.1%. If the density of the water is $1 \times 10^3\ \mathrm{kg/m}^3$ and $\mathrm{g} = 10\ \mathrm{m/s}^2$, then the volume elasticity is	(s) $2 \times 10^9\ \mathrm{N/m}^2$

(A) A-(p) B-(q) C-(r) D-(s)

(B) A-(p) B-(q) C-(s) D-(r)

(C) A-(q) B-(p) C-(r) D-(s)

(D) A-(q) B-(p) C-(s) D-(r)

Q.19

A material breaks up under a stress of $20 \times 10^5\ \mathrm{N/m}^2$. If the density of the material is $2.5 \times 10^3\ \mathrm{kg/m}^3$, the length (in m) may break under its own weight is 10 times x. Find the value of x.(take $g = 10\ \mathrm{m/s}^2$).

Q.20

A uniform rod of mass m and length I is rotating with constant angular yelocity ω about an axis which passes through its one end and perpendicular to the length of rod. The area of cross section of the rod is A and its young's modulus is Y. Neglect gravity. The strain at the midpoint of the rod is $\frac{3m\omega^2 1}{KAY}$. Then the value of ' K ' is

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Answers & Solutions

Q.1 Answer:

$\frac{2L}{3}$

Solution:

Let T_A and T_B be the tensions in wires A and B respectively. If a_A and a_A are the respective cross sectional areas, then

Stress in wire $A = \frac{T_A}{a_A}$

Stress in wire $B = \frac{T_B}{a_B}$

The stress in wires A and B will be equal is

$$\frac{T_A}{a_A} = \frac{T_B}{a_B} \text{ or } \frac{T_A}{T_B} = \frac{a_A}{a_B} = \frac{1}{2}$$

Since the system is in equilibrium, the moments of forces T_A and T_B about C will be equal

$$T_A \times x = T_B \times (L - x)$$
$$\text{Or } \frac{T_A}{T_B} = \frac{L - x}{x} \text{ or } \frac{1}{2} = \frac{L - x}{x}$$
$$\therefore x = \frac{2L}{3}$$

Q.2 Answer:

88 N

Solution:

$\mathrm{F} = \mathrm{Y}\alpha\Delta\mathrm{tA}$

$\mathrm{A} = 2 \times 10^{-6}\ \mathrm{m}^2, \mathrm{Y} = 2 \times 11\ \mathrm{N/m}^2$

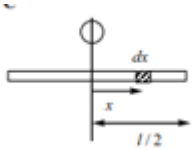
$\alpha = 1.1 \times 10^{-5}, \mathrm{t} = 50 - 30 = 20^{\circ}\mathrm{C}$

$\mathrm{F} = 2 \times 10^{11} \times 1.1 \times 10^{-5} \times 20 \times 2 \times 10^{-6} = 88\ \mathrm{N}.$

Q.3 Answer:

$\sqrt{\frac{2\sigma}{\rho}}\frac{1}{\pi l}$

Solution:



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$$-dT = (A\rho)(dx)x\omega^2$$
$$-dT = A\rho\omega^2 x(dx)$$
$$-(T)_O^T = \frac{A\rho\omega^2}{2}\left(x^2\right)_{l/2}^2$$
$$T = \frac{A\rho\omega^2}{2}\frac{l^2}{4}$$
$$\Rightarrow \frac{T}{A} = \frac{\rho\omega^2 l^2}{8}$$
$$\Rightarrow \frac{\rho\omega^2 l^2}{8} < \sigma$$
$$\omega^2 < \frac{8\sigma}{\rho l^2}$$
$$\Rightarrow 4\pi^2 f^2 < \frac{8\sigma}{\rho l^2}$$
$$\Rightarrow f^2 < \frac{2\omega}{\rho}\frac{1}{\pi^2 l^2}$$
$$f_{Max} = \sqrt{\frac{2\sigma}{\rho}}\frac{1}{\pi l}$$

Q.4 Answer:

8.1×10^6 ergs

Solution:

W $= \frac{1}{2} \times$ load $\times$ **elongation**
W $= \frac{1}{2} \times 5.4 \times 10^6 \times 3 = 8.1 \times 10^6$ **ergs**

Q.5 Answer:

$\frac{I_1T_2 - I_2T_1}{T_2 - T_1}$

Solution:

$Y = \frac{Fl}{a\Delta}$
Y, l **and** a **are constant**
$\therefore \frac{FI}{\Delta l} =$ **constant or** $\Delta = F$
Now, $i_1 - i = T_1$ **and** $l_2 - l_1 = T_2$
Diving, $\frac{I_1 - l}{I_2 - l} = \frac{T_1}{T_2}$
or $l_1T_2 - IT_2 = l_2T_1 - IT_1$ **or** $I = \frac{I_1T_2 - I_2T_1}{T_2 - T_1}$

Q.6 Answer:

2×10^9 Pa

Solution:

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$\mathrm{B} = -\frac{\Delta \mathrm{PV}}{\Delta \mathrm{V}}$

Given, $\Delta \mathrm{P} = h\rho g = 200 \times 10^3 \times 10$

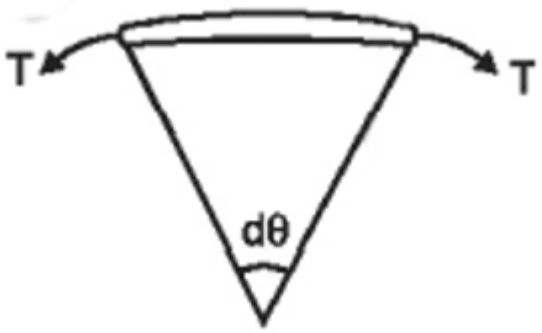
$\Delta \mathrm{P} = 2 \times 10^6 \ \mathrm{N/m^2}$

$\frac{\Delta \mathrm{V}}{\mathrm{V}} = \frac{0.1}{100} = 10^{-3} \quad \therefore \mathrm{B} = \frac{2 \times 10^6}{10^{-3}} = 2 \times 10^9 \ \mathrm{N/m^2}$

Q.7 Answer:

$dR = \frac{\rho \omega^2 R^3}{Y}$

Solution:



$$
\begin{aligned}
& 2\ \mathrm{T}.\ \sin\frac{\mathrm{d}\theta}{2} = \mathrm{dm}\omega^2\mathrm{R} \\
\Rightarrow \quad & \mathrm{T}.\mathrm{d}\,\theta = \rho A \mathrm{A} \mathrm{r} \mathrm{d}\theta \omega^2 \mathrm{R} \\
\Rightarrow \quad & \mathrm{T} = \rho A R^2 \omega^2 \\
\therefore \quad & \frac{\mathrm{dR}}{\mathrm{R}} = \frac{\mathrm{T}}{\mathrm{AY}} \Rightarrow \quad \Rightarrow \quad \mathrm{dR} = \frac{\rho A R^3 \omega^2}{\mathrm{AY}} = \frac{\rho \omega^2 \mathrm{R}^3}{\mathrm{Y}}.
\end{aligned}
$$

Q.8 Answer:

$1.13 \ \mathrm{mm}$

Solution:

Limiting stress $= 4.0 \times 10^8 \ \mathrm{N/m^2}$

$\frac{\mathrm{F}}{\mathrm{A}} = \frac{400}{\mathrm{A}} = 4.0 \times 10^8$ **or** $\mathrm{A} = 10^{-6} \ \mathrm{m^2}$

$\therefore \quad \mathrm{D} = \left(\frac{4\,\mathrm{A}}{\pi}\right)^{1/2} = \left(\frac{4 \times 10^{-6}}{\pi}\right)^{1/2} = 1.13 \times 10^{-3} \ \mathrm{m} = 1.13 \ \mathrm{mm}$

Q.9 Answer:

B stretches more than A.

Solution:

According to Equation $[\mathrm{F} = \mathrm{Y}\,(\Delta \mathrm{L}/\mathrm{L}_0)\,\mathrm{A}]$**, the force** F **required to stretch a piece of material is proportional to Young's modulus** Y**, the amount of stretch** ΔL**, and the cross-sectional area** A **of the material, but is inversely proportional to the initial length** L_0 **of the material.**

Solving this equation for the amount of stretch gives $\Delta \mathrm{L} = \mathrm{FL}_0/\mathrm{YA}$**.**

Thus, the greater the cross-sectional area, the smaller is the amount of stretch, for given values of Young's

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modulus, the initial length, and the stretching force.

Thus, B stretches more than A, because B has the smaller cross-sectional area of solid material.

Q.10 Answer:

$\mathrm{T_{cu}} = 245\ \mathrm{N}$

Solution:

Let tension in copper and steel wires are $\mathrm{T_{cu}}$ and $\mathrm{T_{st}}$.

Assuming same strain (due to heavy plank)

We note that $Y_{\text{steel}} = \frac{\mathrm{T_{st}\,L}}{\mathrm{A(\Delta L)}}$ **and** $Y_{\mathrm{cu}} = \frac{\mathrm{T_{cu}L}}{\mathrm{A(\Delta L)}}$ **or** $\frac{\mathrm{T_{st}}}{\mathrm{T_{cu}}} = \frac{\mathrm{Y_{steel}}}{\mathrm{Y_{cu}}} = 2$ **(given)**

or $\mathrm{T_{st}} = 2\ \mathrm{T_{cu}}$

Further, $\mathrm{T_{st}} + 2\ \mathrm{T_{cu}} = 100\ \mathrm{g}$ **(given)**

or $2\ \mathrm{T_{st}} = 2\ \mathrm{T_{cu}} = 100\ \mathrm{g}$ **or** $\mathrm{T_{cu}} = 25\ \mathrm{g}$

And therefore, $\mathrm{T_{st}} = 2\ \mathrm{T_{cu}} = 50\ \mathrm{g}$

Thus $\mathrm{T_{cu}}$ **is** $25\mathrm{kgf}$

$\mathrm{T_{steel}}$ **is** $50\ \mathrm{kgf}$**, that is** $\mathrm{T_{cu}} = 245\ \mathrm{N}$ **and** $\mathrm{T_{steel}}$ **is** $490\ \mathrm{N}$.

Q.11 Answer:

If a heavy uniform rod is suspended vertically, its diameter is largest at the bottom and smallest at the top.

Solution:

A load decreases from top to bottom.

Therefore, elongation decreases from top to bottom.

$\therefore$ **Lateral strain more at top.**

$\therefore$ **Diameter is least at top.**

(B) $\mathrm{Y} = \frac{\mathrm{F}}{\mathrm{A}(\Delta\ell/\ell)} = \frac{\mathrm{F}}{\mathrm{A}.\,\alpha\theta} \Rightarrow \mathrm{F} = \mathrm{AY}\alpha\theta$ **independent of** ℓ.

(C) Shear modulus zero (cannot withstand shear force)

(D) Infinity. Therefore $\Delta\mathrm{V}/\mathrm{V} = 0$ **whatever be pressure.**

Q.12 Answer:

$\mathrm{mg}\ell$

Solution:

Loss in gravitational $\mathrm{PE} = \mathrm{mgh}$

Gain in elastic P.E. $= \frac{1}{2}\times$ **Load** $\times$ **elongation** $= \frac{1}{2}\mathrm{mg}\ell$

Balance $= \frac{1}{2}\mathrm{mg}\ell =$ **heat produced.**

Q.13 Answer:

$\mathrm{mg}\ell/2$

Solution:

Loss in gravitational $\mathrm{PE} = \mathrm{mgh}$

Gain in elastic P.E. $= \frac{1}{2}\times$ **Load** $\times$ **elongation** $= \frac{1}{2}\mathrm{mg}\ell$

Balance $= \frac{1}{2}\mathrm{mg}\ell =$ **heat produced.**

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Q.14 Answer:

$mg\ell/2$

Solution:

Loss in gravitational $PE = mgh$

Gain in elastic P.E. $= \frac{1}{2} \times$ **Load** $\times$ **elongation** $= \frac{1}{2} mg\ell$

Balance $= \frac{1}{2} mg\ell =$ **heat produced.**

Q.15 Answer:

cannot be concluded

Solution:

(i) Elastic strain energy per unit volume, E

$E = \frac{1}{2} \times \text{ stress } \times \text{ strain } = \frac{1}{2} \times Y \times (\text{ strain })^2 = \frac{1}{2} \times Y \times \frac{(\Delta\ell)^2}{\ell^2}$

$\frac{dE}{d(\Delta\ell)} = \frac{Y}{\ell^2} \cdot \Delta\ell$

Slope at any $\Delta\ell \propto \frac{Y}{\ell^2}$ **cannot be concluded without** ℓ_1, ℓ_2 **data**

Q.16 Answer:

$x < y$

Solution:

(ii) Thermal stress $= \left(\frac{\alpha_1\ell_1 + \alpha_2\ell_2}{\frac{\ell_1}{Y_1} + \frac{\ell_2}{Y_2}} \right) \theta \ldots (1)$

without support : $x = (\alpha_1\ell_1 + \alpha_2\ell_2)\,\Delta\theta$

$y = \frac{\sigma\ell_1}{Y_1} + \frac{\sigma\ell_2}{Y_2}$

$\therefore \quad (1) \Rightarrow$ **thermal stress** $= \frac{x}{y/\sigma} = \frac{x}{y}\sigma$

Given $\frac{x}{y}\sigma < \sigma \Rightarrow \frac{x}{y} < 1 \Rightarrow 1 \Rightarrow x < y$

Q.17 Answer:

0.56

Solution:

(iii) Formula: $Y = \frac{\sigma}{\Delta\ell/\ell} \Rightarrow \Delta\ell = \frac{\sigma \cdot \ell}{Y}$

Clearly $\Delta\ell$ **is same for the three wires (** $\therefore$ **beam remains horizontal)**

Let suffixes be 1 **: steel,** 2 **: brass,** 3 **: composite**

Let f **be fraction of steel in third wire. Then, for third wire**

$\Delta\ell = \frac{\sigma_3 \cdot f \cdot \ell}{Y_1} + \frac{\sigma_3(1-f)}{Y_2}\ell \quad \therefore \frac{\sigma_1\ell}{Y_1} = \frac{\sigma_2\ell}{Y_2} = \sigma_3\ell\left(\frac{f}{Y_1} + \frac{1-f}{Y_2}\right)$

$\Rightarrow \quad \frac{\sigma_1}{20} = \frac{\sigma_2}{8} = \sigma_3\left(\frac{f}{20} + \frac{1-f}{8}\right) \Rightarrow \sigma_2 = 0.4\sigma_1, \sigma_3 = \frac{2}{(5-3f)} \cdot \sigma_1$

Since 6170 is half of 1234 **g, one of the three wires takes half the load. It cannot be wire**

$2\,(\therefore \sigma_2 = 0 \cdot 4\sigma_1 < \sigma_1)$

$\therefore$ **Either,**

$\Rightarrow \sigma_1 + \sigma_2 = \sigma_3$ **or** $\sigma_2 + \sigma_3 = \sigma_1$

$\Rightarrow \quad 1.4\sigma_1 = \frac{2}{5-3f} \cdot \sigma_1$ **or** $0.4\sigma_1 + \frac{2\sigma_1}{5-3f} = \sigma_1$

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$\Rightarrow f > 1$ or $f = \frac{5}{9} = 0.56$

(not possible)

Q.18 Answer:

A-(p) B-(q) C-(r) D-(s)

Solution:

$(A) - \mathbf{p}, (B) - \mathbf{q}, (C) - \mathbf{r}, (D) - \mathbf{s}.$

(A) Here $\Delta P = 100.0 \text{ atm} = 100.0 \times 1.013 \times 10^5 \text{Nm}^{-2}. \Delta V = 100.5 - 100.0 = 0.5$ **litre**

$V = 100.0$ litre

$$\therefore \quad B = \frac{V\Delta P}{\Delta V} = \frac{100.0 \times 1.013 \times 10^7}{0.5} = 2.03 \times 10^9 \text{Nm}^{-2}$$

(B) Here $L_s = 4.7 \text{ m}, \ L_c = 3.5 \text{ m}, a_s = 3.0 \times 10^{-5} \text{ m}^2,$

$a_c = 4.0 \times 10^{-5} \text{ m}^2$

$\Delta L_s = \Delta L_c \quad F_s = F_c$

$Y_s = \frac{F_s L_s}{a_s \Delta L_s}$ and $Y_c = \frac{F_c L_c}{a_c \Delta L_c}$

$\frac{Y_s}{Y_c} = \frac{L_s}{a_s} \cdot \frac{a_c}{L_c} = \frac{4.7}{3.0 \times 10^{-5}} \times \frac{4 \times 10^{-5}}{3.5} = 1.79$

(C) $\frac{dV}{V} = (1 + 2\sigma)\frac{dL}{L}; \frac{dV}{V} = 2 \times 2 \times 10^{-3} = 4 \times 10^{-3} \quad \left[\because \sigma = 0.5 = \frac{1}{2}\right]$

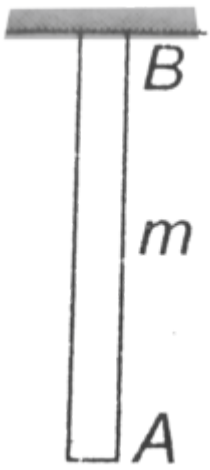
$\therefore$ **Percentage change in volume** $= 4 \times 10^{-1} = 0.4\%$

(D) $K = \frac{\Delta P}{\Delta V/V} = \frac{h\rho g}{\Delta V/V} = \frac{200 \times 10^3 \times 10}{0.1/100} = 2 \times 10^9$

Q.19 Answer:

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Solution:



stress at uppermost point B **is maximum.**

Breaking condition: stress $_B \geq 20 \times 10^5 \text{ N/m}^2$

stress $_B = F/A$**, here** F **is weight of material.**

$\text{stress}_B = mg/A = 20 \times 10^5$

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mass of material is given by $\mathrm{m} = \mathrm{AL}\rho$

$20 \times 10^5 = (\mathrm{ALpg})/\mathrm{A}$

$20 \times 10^5 = \mathrm{Lpg}$

$\mathrm{L} = \frac{20 \times 10^5}{\rho \mathrm{g}}$

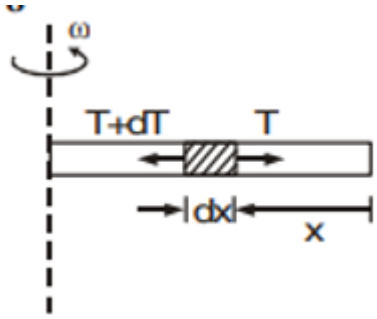
$\mathrm{L} = \frac{20 \times 10^5}{2.5 \times 10^3 \times 10}$

$\mathrm{L} = 80\ \mathrm{m} = 10 \times 8\ \mathrm{m}$. So, $x = 8$

Q.20 Answer:

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Solution:



$dT = dm(1-x)w^2 \quad dT = \frac{m}{1} \cdot dx(1-x)\omega^2$

$\int_0^T dT = \int_0^{1/2} \frac{m\omega^2}{1}(1-x)dx$

$= \frac{m\omega^2}{1}\left[1x - \frac{x^2}{2}\right]_0^{1/2} = \frac{m\omega^2}{1}\left[\frac{1^2}{2} - \frac{1^2}{8}\right]$

Tension at mid-point is :s

$T = \frac{3}{8}ml\omega^2 \Rightarrow \text{ stress } = \frac{3ml\omega^2}{8A} \Rightarrow \text{ strain } = \frac{3ml\omega^2}{8AY}$

So, $k = 8$

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